Synthetic Evaluation of Temporal Grounding

1 MoA Temporal Reconciliation Evaluation

We evaluate a 4-stage Mixture-of-Agents (MoA) temporal grounding pipeline on a controlled synthetic benchmark with known ground-truth year labels. The benchmark has 120 samples: 60 explicit, 20 explicit-multi, 20 implicit, and 20 multi-implicit. [1]

Pipeline. We run three proposer agents (`gemini-3-flash`, `claude-4.5-haiku`, `gpt-5-mini`) and one aggregator (`gemini-3-flash`) in four stages:

- **Proposal round 1:** three independent proposer outputs.
- **Aggregation round 1:** aggregator synthesizes round-1 outputs.
- **Proposal round 2:** three proposers re-propose using the round-1 aggregate as auxiliary context.
- **Aggregation round 2:** aggregator synthesizes round-2 outputs.

Scoring. For each proposal round, exact-match accuracy is averaged over the three proposer predictions per sample (values in $\{0, \frac{1}{3}, \frac{2}{3}, 1\}$), then averaged across samples. Aggregation rounds are scored as standard binary exact-match on the aggregated year. For implicit and multi-implicit subsets, we also report Wasserstein distance (Earth Mover distance; lower is better) between predicted year distributions and the ground-truth point mass at the gold year.

Current results (latest run). Overall, aggregation round 1 is best (88.3%), improving over proposal round 1 (85.0%). After the second proposer pass, proposal round 2 is 83.1%, and final aggregation round 2 is 83.3%. Performance is near ceiling on explicit variants (96.7–100.0%), while implicit remains hardest (35.0–46.7%). Multi-implicit benefits most from early aggregation (85.0% at aggregation round 1). Tables 1 and 2 provide the exact numerical breakdown behind these statements: the first table reports overall stage-wise exact-match,

[1] Explicit: single timestamp target. Explicit-multi: one sentence mentions two entities, and the target is the later entity year. Implicit: one non-event claim with plausible-year uncertainty. Multi-implicit: one sentence combines two implicit claims; target is earliest year where both can hold jointly.

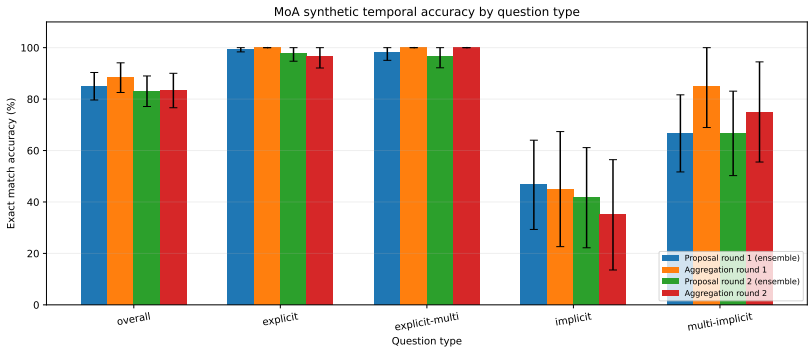

Figure 1: MoA synthetic temporal evaluation by question type with exact-match accuracy and 95% confidence intervals.

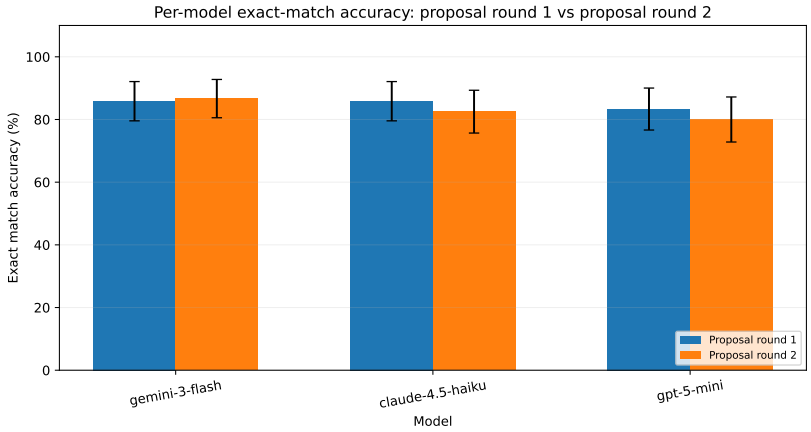

Figure 2: Per-model exact-match accuracy: proposal round 1 vs proposal round 2.

while the second resolves those same stages by question type. Tables 3 and 4 complement exact-match with distributional quality, showing where aggregation helps reduce temporal distance even when strict exact-year accuracy is difficult.

Representative cases

Table 5 shows an explicit sample, Table 6 shows an explicit-multi sample, Table 7 shows an implicit sample with distributional outputs, and Table 8 shows a multi-implicit sample. Table 9 isolates proposal round 1 outputs for the same implicit sentence across all three proposer models. Each example table uses the same row layout (*Gold*, *Input*, proposer outputs, and final aggregation output) so readers can compare how evidence changes between rounds. The intent is to make failure and recovery patterns auditable at the sample level: where round-

Table 1: Overall exact-match accuracy by MoA stage (120 samples).

MoA stage	Accuracy (%)	Correct / Total
Proposal round 1 (ensemble)	85.0	102.0 / 120
Aggregation round 1	88.3	106.0 / 120
Proposal round 2 (ensemble)	83.1	99.7 / 120
Aggregation round 2	83.3	100.0 / 120

Table 2: Exact-match accuracy by question type and MoA stage.

Question type	Proposal 1	Aggregation 1	Proposal 2	Aggregation 2
Explicit	99.4	100.0	97.8	96.7
Explicit-multi	98.3	100.0	96.7	100.0
Implicit	46.7	45.0	41.7	35.0
Multi-implicit	66.7	85.0	66.7	75.0

2 proposals become more consistent, where aggregation corrects disagreement, and where the pipeline still drifts.

2 Related Work

Our setup follows layered multi-agent synthesis in MoA pipelines, where parallel first-pass generations are iteratively aggregated and refined [3]. For exact-label tasks, self-consistency and voting remain strong controls [5, 4]. The refinement stage is also related to critique/refine and debate-style interaction [2, 1].

3 Next Steps

1. Improve implicit calibration (especially implicit stage-4 regression) via stronger distribution constraints and calibration-aware aggregation.
2. Compare this fixed MoA schedule against debate-style proposer interactions while keeping the same final aggregation model [1].

References

[1] Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate, 2023.

[2] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-refine: Iterative refinement with self-feedback, 2023.

Table 3: Implicit-only Wasserstein distance by MoA stage (lower is better).

MoA stage	Median distance (years)	Count
Proposal round 1	2.122	20
Aggregation round 1	0.550	20
Proposal round 2	1.500	20
Aggregation round 2	1.400	20

Table 4: Multi-implicit Wasserstein distance by MoA stage (lower is better).

MoA stage	Median distance (years)	Count
Proposal round 1	0.717	20
Aggregation round 1	0.350	20
Proposal round 2	0.458	20
Aggregation round 2	0.325	20

[3] Junlin Wang, Jue Wang, Ben Athiwaratkun, Ce Zhang, and James Zou. Mixture-of-agents enhances large language model capabilities. In *International Conference on Learning Representations (ICLR)*, 2025.

[4] Weiqin Wang, Yile Wang, and Hui Huang. Ranked voting based self-consistency of large language models. In *Findings of the Association for Computational Linguistics: ACL 2025*, 2025.

[5] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models, 2022.

Table 5: Explicit sample (id 103), with `gpt-5-mini` proposer outputs and `gemini-3-flash` aggregation output.

Row label	Text
Gold	2014
Input	"Google announces the release of Kubernetes as an open-source container orchestration tool."
Proposal round 1 (`gpt-5-mini`)	"Google open-sourced Kubernetes and announced the project in mid-2014 (the code was published on GitHub in June 2014). The 1.0 stable release came later (July 2015), but the public open-source announcement occurred in 2014. Final year: 2014."
Proposal round 2 (`gpt-5-mini`)	"Kubernetes was publicly announced and released as an open-source project by Google in mid-2014 (announced at DockerCon in June 2014). The first stable 1.0 release came later in July 2015, but the initial public open-source release occurred in 2014. Final year: 2014."
Aggregation round 2 (`gemini-3-flash`)	"Google officially announced Kubernetes as an open-source project in June 2014 at DockerCon. While the first stable version (1.0) was released in July 2015, the initial public announcement and open-source code release occurred in 2014. Final year: 2014."

Table 6: Explicit-multi sample (id 62), with `gpt-5-mini` proposer outputs and `gemini-3-flash` aggregation output.

Row label	Text
Gold	2001
Input	"The Lord of the Rings: The Fellowship of the Ring premiered in theaters the same year as Harry Potter and the Sorcerer's Stone."
Proposal round 1 (`gpt-5-mini`)	"The Fellowship of the Ring premiered in December 2001 and Harry Potter and the Sorcerer's Stone premiered in November 2001; both films premiered in 2001, so the later (latest) year is 2001. Final year: 2001."
Proposal round 2 (`gpt-5-mini`)	"Both films had theatrical premieres in 2001: Harry Potter in November and Fellowship in December. Since both premieres occurred in the same calendar year, the latest year is 2001. Final year: 2001."
Aggregation round 2 (`gemini-3-flash`)	"The Fellowship of the Ring premiered in December 2001 and Harry Potter and the Sorcerer's Stone premiered in November 2001. Because both were released in the same calendar year, the latest year is 2001. Final year: 2001."

Table 7: Implicit sample (id 1), with `gpt-5-mini` proposer outputs and `gemini-3-flash` aggregation output.

Row label	Text
Gold	"1964. Gold plausible years: [1964,1965,1966,1967,1968]. Gold plausible probs: [0.50,0.20,0.10,0.10,0.10]."
Input	"Federal health authorities have officially concluded that cigarette smoking is a primary cause of lung cancer and chronic bronchitis."
Proposal round 1 (`gpt-5-mini`)	"The U.S. Surgeon General's 1964 report *Smoking and Health* is the canonical federal declaration concluding cigarette smoking is a cause of lung cancer and chronic bronchitis; this is the earliest clear official federal conclusion. Final year: 1964."
Proposal round 2 (`gpt-5-mini`)	"The earliest clear official federal conclusion is the U.S. Surgeon General's report published on January 11, 1964. Earlier studies provided strong evidence, but 1964 is the first formal federal health authority statement of this kind. Final year: 1964."
Aggregation round 2 (`gemini-3-flash`)	"This statement aligns with the landmark Surgeon General's Report on Smoking and Health released on January 11, 1964. Earlier evidence existed, but 1964 is when a U.S. federal health authority issued a definitive official conclusion. Plausible years: [1964]. Plausible probs: [1.0]. Final year: 1964."

Table 8: Multi-implicit sample (id 115), with `gpt-5-mini` proposer outputs and `gemini-3-flash` aggregation output.

Row label	Text
Gold	"2020. Gold plausible years: [2020,2021,2022,2023]. Gold plausible probs: [0.50,0.30,0.10,0.10]."
Input	"Volunteer groups focused on filtering microplastics from ocean waters have modified their operations to ensure strict adherence to social distancing mandates."
Proposal round 1 (`gpt-5-mini`)	"The phrase 'social distancing mandates' is most plausibly tied to the COVID-19 pandemic, when such mandates were widely implemented beginning in March 2020. Volunteer groups modified operations throughout 2020 and into 2021, so 2020 is the earliest reasonable year. Final year: 2020."
Proposal round 2 (`gpt-5-mini`)	"Widespread social distancing mandates were first implemented in response to COVID-19 in early 2020. Operational changes to comply with those mandates are therefore most plausibly first observed in 2020, with continuation into 2021 possible. Final year: 2020."
Aggregation round 2 (`gemini-3-flash`)	"The phrase 'social distancing mandates' identifies COVID-19 as the temporal anchor. Those mandates began in early 2020, so this specific operational shift could first occur in 2020. Plausible years: [2020,2021]. Plausible probs: [0.90,0.10]. Final year: 2020."

Table 9: Implicit sample (id 1): proposal round 1 outputs across all three proposer models.

Row label	Text
Input	"Federal health authorities have officially concluded that cigarette smoking is a primary cause of lung cancer and chronic bronchitis."
Proposal round 1 (`gemini-3-flash`)	"The landmark U.S. Surgeon General's Report on Smoking and Health (January 11, 1964) marked the first official federal conclusion establishing smoking as a primary cause of lung cancer and chronic bronchitis. Plausible years: [1964,1965]. Plausible probs: [0.95,0.05]. Final year: 1964."
Proposal round 1 (`claude-4.5-haiku`)	"The 1964 Surgeon General's report was the first major U.S. federal health authority document officially concluding smoking causes lung cancer and chronic bronchitis. Plausible years: [1964,1965,1966]. Plausible probs: [0.75,0.20,0.05]. Final year: 1964."
Proposal round 1 (`gpt-5-mini`)	"The U.S. Surgeon General's 1964 report *Smoking and Health* is the canonical federal conclusion linking smoking to lung cancer and chronic bronchitis. Plausible years: [1964]. Plausible probs: [1.0]. Final year: 1964."